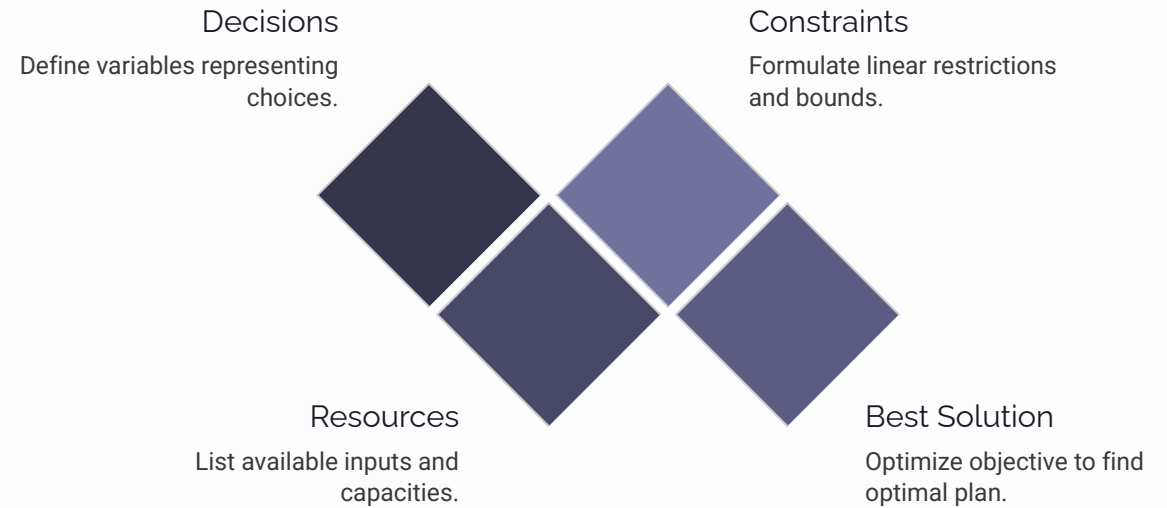


Linear Programming

Optimization Under Constraints



Start With a Real Question

"You have limited money, time, and computing power — how do you get the most out of them?"

Limited GPU Hours

Training deep models is expensive. You can't run everything.

Limited Cloud Budget

Every API call and compute instance costs money.

Limited Employee Time

Your team can't work on every project at once.

Limited Raw Materials

Factories can't produce more than their supply allows.

 **Linear Programming helps us make the best possible decision under limitations.**



What is Linear Programming?

Linear Programming (LP) is a mathematical technique used to **maximize or minimize** a linear objective function, subject to a set of linear constraints.



Decision Variables

Objective Function

Constraints

Feasible Solutions

Think of LP as a structured decision-making recipe: you define **what you're deciding**, **what you want**, and **what's limiting you** — and the method finds the best answer that respects all the rules.

The Four Building Blocks of LP

1

Decision Variables

What are we deciding?

$x = \text{Units of Product A}, \quad y = \text{Units of Product B}$

2

Objective Function

What do we want to optimize?

$$Z = 40x + 30y$$

3

Constraints

What limits our decision?

$$2x + y \leq 100$$

4

Non-Negativity

Values must be realistic.

$$x, y \geq 0$$

📌 **Key Learning:** Every LP problem – no matter how complex – is built from exactly these four components.

Decision Variables Explained

Decision variables are the **unknowns we want to find**. They represent the quantities our decision-maker controls.

Ask yourself: "**What are we trying to decide?**"

x = Number of Laptops

y = Number of Tablets

Once solved, x and y will tell you exactly how many of each to produce for the best outcome.



 x = Laptops

The number of laptops to manufacture each day.

 y = Tablets

The number of tablets to manufacture each day.

The Objective Function

The objective function tells us **what we want to optimize** — it's the goal we're chasing. It's always a linear expression of the decision variables.

Example: Laptop profit = ₹5,000 | Tablet profit = ₹3,000

$$\text{Maximize } Z = 5000x + 3000y$$

 Maximize

Profit, Production, Research Value, Experiments

 Minimize

Cost, Transportation Cost, Time, Waste



Constraints — The Rules You Cannot Break



Real Constraint Example

A **laptop** requires **4 hours** of assembly.

A **tablet** requires **2 hours** of assembly.

Total available time = **40 hours**.

$$4x + 2y \leq 40$$

In plain English: *"The total assembly time used cannot exceed 40 hours."*

Constraints can use $\leq$ (at most), $\geq$ (at least), or $=$ (exactly). Most resource constraints use $\leq$.

Non-Negativity Constraints

In most real-world problems, **negative quantities make no physical sense**. You cannot produce -5 laptops or run -3 experiments.

$$x \geq 0, \quad y \geq 0$$

This simple rule ensures the mathematical solution stays **practically meaningful**. Without it, solvers might return absurd negative values that technically optimize Z but are useless in reality.

If $x = -5$ laptops, what does that mean? Nothing useful. Non-negativity prevents this.

📌 **Key Learning:** Non-negativity keeps the math grounded in the real world.



The Complete LP Standard Form

Here is a complete LP problem. Every part is labeled – learn to recognize this structure instantly.

 Objective Function

$$\text{Maximize } Z = 40x + 30y$$

 Constraint 1

$$2x + y \leq 100$$

 Constraint 2

$$x + 2y \leq 80$$

 Non-Negativity

$$x, y \geq 0$$

Practice writing LP problems in this exact format. It makes solutions much easier to check and verify.

Feasible vs. Infeasible Solutions

✓ FEASIBLE

Point: $(x, y) = (20, 30)$

Check against constraint $2x + y \leq 100$:

$$2(20) + 30 = 70 \leq 100 \quad \checkmark$$

This point **satisfies** the constraint — it's inside the allowed region. A fully feasible solution must satisfy **every** constraint simultaneously.

✗ INFEASIBLE

Point: $(x, y) = (40, 30)$

Check against constraint $2x + y \leq 100$:

$$2(40) + 30 = 110 > 100 \quad \text{⚡}$$

This point **violates** the constraint — it exceeds available resources. Even one violation makes the entire solution infeasible.

📌 **Key Learning:** Feasible = all constraints satisfied. Infeasible = at least one constraint violated.

The Feasible Region



What is the Feasible Region?

The **feasible region** is the set of *all* points that satisfy every constraint simultaneously.

For the system:

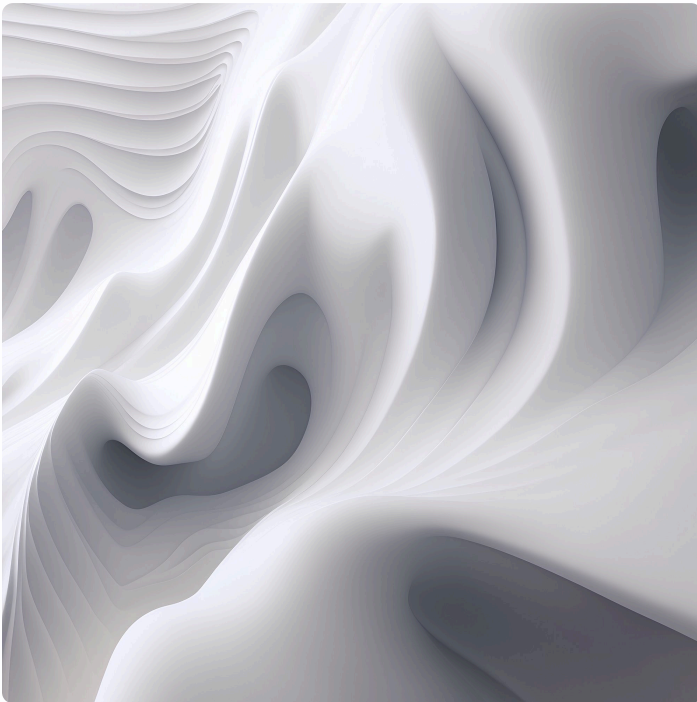
$$2x + y \leq 100$$

$$x + 2y \leq 80$$

$$x, y \geq 0$$

The shaded area on the graph shows every allowed (x, y) combination. The **optimal solution** always lives somewhere in — or on the boundary of — this region.

Real-World Analogy: The Playground



Think of it This Way...



Fences = Constraints

The boundaries you cannot cross, no matter what.



Playground = Feasible Region

The safe zone where all your options live.



Any Point Inside = Feasible Solution

You're allowed to stand anywhere inside the fence.



Outside = Infeasible

You've crossed the fence — not allowed!

Corner Points & the Graphical Method

Why Corner Points?

In any two-variable LP problem, the **optimal solution always occurs at a corner point** (vertex) of the feasible region — never at a random interior point.

This means instead of checking *infinitely many* points, we only evaluate the **small, finite set of corner points**. Much more manageable!

01

Draw constraint lines on a graph

02

Identify the shaded feasible region

03

Find all corner points (vertices)

04

Evaluate Z at each corner point

05

Pick the best feasible value of Z



Complete Graphical Example — Step by Step

Maximize $Z = 40x + 30y$ subject to $2x + y \leq 100$, $x + 2y \leq 80$, $x, y \geq 0$



Step 1: Find Corner Points

Intersect boundary lines + axes to get candidate vertices.

Step 2: Verify Feasibility

Confirm each point satisfies *all* constraints.



Step 3: Evaluate Z

Substitute each corner point into $Z = 40x + 30y$.

Step 4: Select Optimal

Choose the feasible point giving the highest Z .

Corner Point	$Z = 40x + 30y$	Feasible?
(0, 0)	0	✔ Yes
(50, 0)	2000	✔ Yes
(0, 40)	1200	✔ Yes
(40, 20) ← Optimal	2200 ✨	✔ Yes

Finding the Intersection Point

Solving the Two Equations

To find the interior corner point, solve the two constraint equations simultaneously:

$$2x + y = 100 \quad \dots(1)$$

$$x + 2y = 80 \quad \dots(2)$$

From (1): $y = 100 - 2x$

Substitute into (2):

$$x + 2(100 - 2x) = 80$$

$$x + 200 - 4x = 80 \implies -3x = -120$$

$$x = 40, \quad y = 100 - 2(40) = 20$$

Result

$$(x, y) = (40, 20)$$

This is the **interior corner point** where both constraint lines meet. Since it satisfies all constraints, it is **feasible**.

$$Z = 40(40) + 30(20) = 1600 + 600 = 2200$$

✓ **Optimal Solution: Z = 2200 at (40, 20)**

LP in the Real World — AI/ML Applications



GPU & Resource Allocation

Assign compute resources across multiple AI jobs to maximize total output value.



Training Scheduling

Decide how to split time between data preprocessing and model training.



Cloud Budget Optimization

Maximize experiments or training runs within a fixed cloud spending limit.



Operations Research

Minimize delivery costs, transportation routes, and logistics under capacity constraints.

Worked Examples — LP Formulations

EXAMPLE 1: CLOUD BUDGET

A student has **₹1,000** for cloud computing. Model A costs **₹100/experiment**, Model B costs **₹200/experiment**. Goal: maximize total experiments.

x = Model A experiments, y = Model B experiments

$$\text{Maximize } Z = x + y$$

$$100x + 200y \leq 1000 \implies x + 2y \leq 10$$

$$x, y \geq 0$$

EXAMPLE 2: GPU ALLOCATION

Job A: **2 GPU-hours**, 5 value units. Job B: **4 GPU-hours**, 8 value units.
Available: **40 GPU-hours**.

x = Job A runs, y = Job B runs

$$\text{Maximize } Z = 5x + 8y$$

$$2x + 4y \leq 40$$

$$x, y \geq 0$$

More Worked Examples

EXAMPLE 3

ML Project Time

10 hours total. Preprocessing: 4 value/hr. Training: 6 value/hr.

$$\text{Max } Z = 4x + 6y$$

$$x + y \leq 10, \quad x, y \geq 0$$

EXAMPLE 4

AI Model Training

Model A: 3 GPU-hrs, 8 perf units. Model B: 2 GPU-hrs, 5 units. GPU ≤ 30 .

$$\text{Max } Z = 8x + 5y$$

$$3x + 2y \leq 30, \quad 2x + y \leq 20$$

EXAMPLE 5

Delivery Minimization

Small: 2 tons, ₹1,000. Large: 5 tons, ₹2,000. Need ≥ 20 tons.

$$\text{Min } Z = 1000x + 2000y$$

$$2x + 5y \geq 20, \quad x, y \geq 0$$

EXAMPLE 6

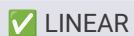
Server Capacity

Workload A: 4 GB RAM, 10 value. Workload B: 2 GB RAM, 6 value. RAM ≤ 24 GB.

$$\text{Max } Z = 10x + 6y$$

$$4x + 2y \leq 24, \quad 3x + 2y \leq 18$$

Linear vs. Nonlinear Optimization



LINEAR

LP requires linear expressions

$$Z = 3x + 5y \quad (\text{Linear})$$

$$2x + y \leq 100 \quad (\text{Linear constraint})$$

Every term is a **constant × variable**. No powers, no products of variables.
This is what makes LP tractable and fast to solve.



NONLINEAR

These are NOT LP problems

$$Z = x^2 + y^2 \quad (\text{Nonlinear — squared term})$$

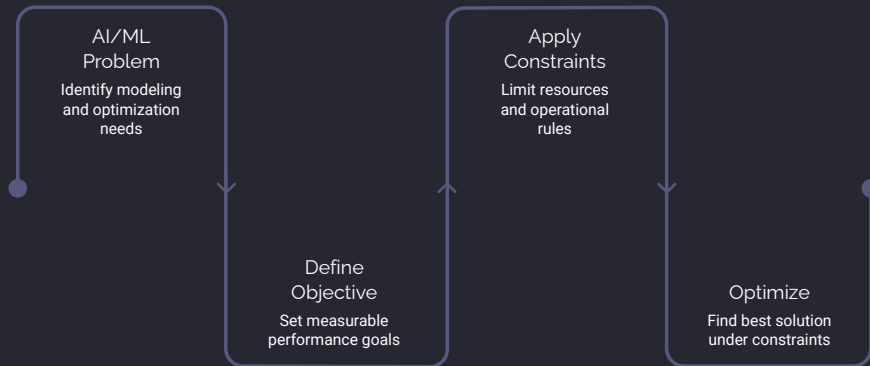
$$Z = xy \quad (\text{Nonlinear — product of variables})$$

Nonlinear problems require different solvers (gradient descent, genetic algorithms, etc.) and are generally **harder to solve**.

📌 **Key Rule:** If your objective or any constraint has squared terms, products of variables, or other nonlinear expressions — it is NOT a Linear Programming problem.

LP and AI/ML — The Connection

Optimization is the Heart of AI/ML



LP is one of the oldest and most reliable optimization techniques — and it still powers real AI/ML systems today.

- Resource Allocation

Distributing GPU/CPU across multiple training jobs.

- Scheduling

Assigning tasks to workers, machines, or servers.

- Feature Selection

Some feature selection problems can be posed as LP relaxations.

- Portfolio Optimization

Allocating budgets across data science projects or experiments.

- Operations Research

Transportation routing, supply chains, network flows.

Note: Not every ML problem is an LP problem. LP is one powerful tool in the optimization toolbox.

Common Student Mistakes to Avoid

✗ Forgetting Variables

Always define x and y clearly before writing anything else.

✗ Wrong Inequality

Reversing $\leq$ and $\geq$ completely changes the feasible region.

✗ Missing Non-Negativity

Always include $x, y \geq 0$ – it's part of the formulation.

✗ Checking One Constraint

A feasible solution must satisfy ALL constraints, not just one.

✗ Skipping Corner Points

Don't guess – evaluate Z at every feasible corner point.

✗ LP = Always Maximize

LP handles both maximization AND minimization problems.

Quick Revision — Key Terms

Term	Meaning	Example
Decision Variable	The quantity we need to determine	x = no. of laptops
Objective Function	The expression we maximize or minimize	$Z = 5000x + 3000y$
Constraint	A rule or limitation on the variables	$4x + 2y \leq 40$
Non-Negativity	Variables must be zero or positive	$x, y \geq 0$
Feasible Solution	A point satisfying all constraints	(20, 30)
Feasible Region	The set of all feasible solutions	Shaded area on graph
Corner Point	A vertex of the feasible region polygon	(40, 20)
Optimal Solution	The feasible point giving best Z value	$Z = 2200$ at (40, 20)

Classroom Practice Problems

For each problem below, identify: Decision Variables · Objective Function · Constraints · Non-Negativity

1	EASY Cloud Budget ₹1,000 budget. Model A: ₹100/run. Model B: ₹200/run. Maximize total experiments.
2	EASY Laptop & Tablet Laptop: ₹5,000 profit, 4 hrs. Tablet: ₹3,000 profit, 2 hrs. Time limit: 40 hrs. Maximize profit.
3	MEDIUM GPU Allocation Job A: 2 GPU-hrs, 5 value. Job B: 4 GPU-hrs, 8 value. Total GPU: 40. Maximize value.
4	MEDIUM Server Capacity Workload A: 4 GB, 10 value. Workload B: 2 GB, 6 value. RAM $\leq$ 24 GB, CPU: $3x+2y \leq 18$.
5	MEDIUM Delivery Cost Small: 2 tons, ₹1,000. Large: 5 tons, ₹2,000. Must carry $\geq$ 20 tons. Minimize cost.
6	MEDIUM AI Research Project A: 2 GPU-hrs, 1 researcher-hr, 8 value. Project B: 1 GPU-hr, 2 researcher-hrs, 10 value. GPU $\leq$ 40, Researchers $\leq$ 50.

Practice Answers

PROBLEM 1: CLOUD BUDGET

$$\text{Max } Z = x + y$$

$$x + 2y \leq 10, \quad x, y \geq 0$$

PROBLEM 2: LAPTOP & TABLET

$$\text{Max } Z = 5000x + 3000y$$

$$4x + 2y \leq 40, \quad x, y \geq 0$$

Step-by-step: Corner points: (0,0) → Z=0; (10,0) → Z=50,000; (0,20) → Z=60,000. **Optimal: (0,20), Z=₹60,000.**

PROBLEM 3: GPU ALLOCATION

$$\text{Max } Z = 5x + 8y$$

$$2x + 4y \leq 40 \implies x + 2y \leq 20, \quad x, y \geq 0$$

PROBLEM 4: SERVER CAPACITY

$$\text{Max } Z = 10x + 6y$$

$$4x + 2y \leq 24, \quad 3x + 2y \leq 18, \quad x, y \geq 0$$

PROBLEM 5: DELIVERY (MINIMIZE)

$$\text{Min } Z = 1000x + 2000y$$

$$2x + 5y \geq 20, \quad x, y \geq 0$$

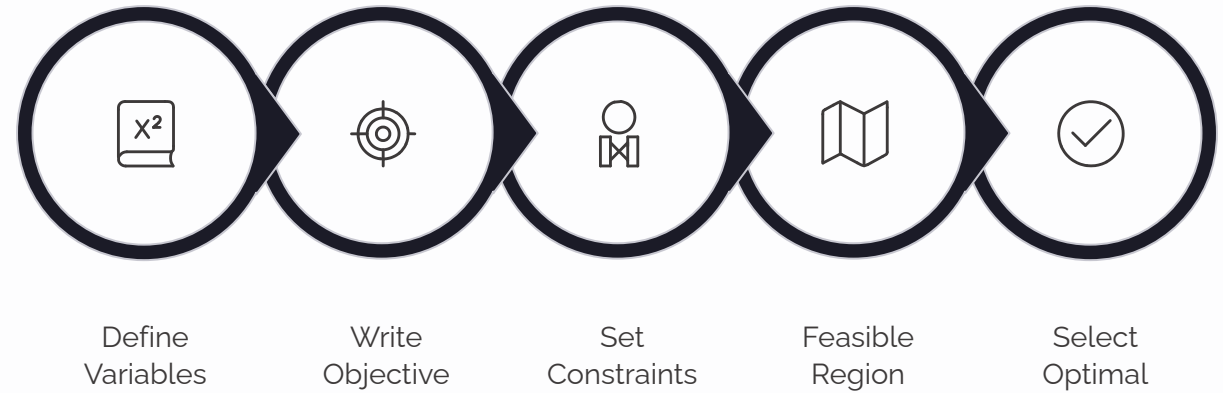
Step-by-step: At (10,0): Z=10,000; (0,4): Z=8,000. **Optimal: (0,4), Z=₹8,000.**

PROBLEM 6: AI RESEARCH

$$\text{Max } Z = 8x + 10y$$

$$2x + y \leq 40, \quad x + 2y \leq 50, \quad x, y \geq 0$$

Final Takeaways



Linear Programming answers one powerful question: "Given limited resources and restrictions, what decision gives the best possible result?"



For AI/ML Students

Limited GPU + Limited Budget +
Limited Time → LP → Better Resource
Decisions



The 4 Components

Decision Variables · Objective ·
Constraints · Non-Negativity — always
present in every LP.



The Method

Graph → Feasible Region → Corner Points → Evaluate Z → Pick Optimal.